

sufficiently many examples they span the whole space, and there will be no zero eigenvalues. This can only happen for $p \geq N$.

When $\lambda = 0$ the solution to (10) inside V_p is just a simple exponential decay to $v_r = 0$. Outside the pattern subspace $A_r = 0$, and the corresponding part of v_r will be constant. Any weight vector which has the same projection onto the pattern subspace as u gives a learning error 0. One can think of this as a 'valley' in the error surface given by $u + V_p^\perp$.

The training set contains no information that can help us choose between all these solutions to the learning problem. When learning with a weight decay $\lambda > 0$, the constant part in $V_p^\perp$ will decay to zero asymptotically (as $e^{-\lambda t}$, where t is the time). An infinitesimal weight decay will therefore choose the solution with the smallest norm out of all the solutions in the valley described above. This solution can be shown to be the optimal one on average.

4 LEARNING WITH AN UNRELIABLE TEACHER

Random errors made by the teacher can be modeled by adding a random term η to the targets:

$$f_u(\xi^\mu) \longrightarrow f_u(\xi^\mu) + \eta^\mu. \quad (12)$$

The variance of η is called σ^2 , and it is assumed to have zero mean. Note that these targets are not exactly realizable by the network (for $\alpha > 0$), and therefore this is a simple model for studying learning of an unrealizable function.

With this noise the learning equation (2) becomes

$$\dot{w}_i \propto \sum_{\mu} (N^{-1} \sum_j v_j \xi_j^\mu + N^{-1/2} \eta^\mu) \xi_i^\mu - \lambda w_i. \quad (13)$$

Transforming it to the basis where $\mathbf{A}$ is diagonal as before,

$$\dot{v}_r \propto -(A_r + \lambda) v_r + \lambda u_r - N^{-1/2} \sum_{\mu} \eta^\mu \xi_r^\mu. \quad (14)$$

The asymptotic solution to this equation is

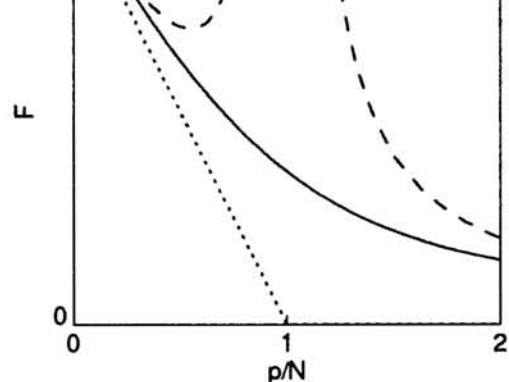
$$v_r = \frac{\lambda u_r - N^{-1/2} \sum_{\mu} \eta^\mu \xi_r^\mu}{\lambda + A_r}. \quad (15)$$

The contribution to the generalization error is the square of this summed over all r . If averaged over the noise (shown by the bar) it becomes for each r

$$F_r = \overline{v_r^2} = \frac{\lambda^2 u_r^2 + N^{-1} \sum_{\mu} (\xi_r^\mu)^2 \overline{(\eta^\mu)^2}}{(\lambda + A_r)^2} = \frac{\lambda^2 u_r^2 + A_r \sigma^2}{(\lambda + A_r)^2}. \quad (16)$$

The last expression has a minimum in λ , which can be found by putting the derivative with respect to λ equal to zero, $\lambda_{\text{optimal}}^r = \sigma^2 / u_r^2$. Remarkably it depends only

for $\lambda = 0$ and $\sigma^2 = 0.2$, and the dashed line for $\lambda = 0$. The dotted line is the generalization error with no noise and $\lambda = 0$.



on $\mathbf{u}$ and the variance of the noise, and *not* on $\mathbf{A}$. If it is assumed that $\mathbf{u}$ is random (16) can be averaged over $\mathbf{u}$. This yields an optimal λ independent of r ,

$$\lambda_{\text{optimal}} = \frac{\sigma^2}{u^2}, \quad (17)$$

where u^2 is the average of $N^{-1}|\mathbf{u}|^2$.

In this case the weight decay to some extent prevents the network from fitting the noise.

From equation (14) one can see that the noise is projected onto the pattern subspace. Therefore the contribution to the generalization error from $V_p^\perp$ is the same as before, and this contribution is on average minimized by a weight decay of any size.

Equation (17) was derived in [10] in the context of a particular eigenvalue spectrum. Figure fig. 1 shows the dramatic improvement in generalization error when the optimal weight decay is used in this case. The present treatment shows that (17) is independent of the spectrum of $\mathbf{A}$.

We conclude that a weight decay has two positive effects on generalization in a linear network: 1) It suppresses any irrelevant components of the weight vector by choosing the smallest vector that solves the learning problem. 2) If the size is chosen right, it can suppress some of the effect of static noise on the targets.

5 NON-LINEAR NETWORKS

It is not possible to analyze a general non-linear network exactly, as done above for the linear case. By a local linearization, it is however, possible to draw some interesting conclusions from the results in the previous section.

Assume the function is realizable, $f = f_u$. Then learning corresponds to solving the p equations

$$f_w(\xi^\mu) = f_u(\xi^\mu) \quad (18)$$

manifold gives a learning error of zero, and therefore (4) can be expanded around $\tilde{\mathbf{w}}$. Putting $\mathbf{v} = \tilde{\mathbf{w}} - \mathbf{w}$, expanding f_w in $\mathbf{v}$, and using it in (4) yields

$$\begin{aligned}\dot{v}_i &\propto - \sum_{\mu,j} \left(\frac{\partial f_w(\xi^\mu)}{\partial w_j} \right) v_j \frac{\partial f_w(\xi^\mu)}{\partial w_i} + \lambda(\tilde{w}_i - v_i) \\ &= - \sum_j \mathcal{A}_{ij}(\tilde{\mathbf{w}}) v_j - \lambda v_i + \lambda \tilde{w}_i\end{aligned}\tag{19}$$

(The derivatives in this equation should be taken at $\tilde{\mathbf{w}}$.)

The analogue of $\mathbf{A}$ is defined as

$$\mathcal{A}_{ij}(\tilde{\mathbf{w}}) \equiv \sum_{\mu} \frac{\partial f_w(\xi^\mu)}{\partial w_i} \frac{\partial f_w(\xi^\mu)}{\partial w_j}.\tag{20}$$

Since it is of outer product form (like $\mathbf{A}$) its rank $R(\tilde{\mathbf{w}}) \leq \min\{p, W\}$. Thus when $p < W$, $\mathcal{A}$ is *never* of full rank. The rank of $\mathcal{A}$ is of course equal to W minus the dimension of the manifold mentioned above.

From these simple observations one can argue that good generalization should *not* be expected for $p < W$. This is in accordance with other results (cf. [3]), and with current ‘folk-lore’. The difference from the linear case is that the ‘rain gutter’ need not be (and most probably is not) linear, but curved in this case. There may in fact be other valleys or rain gutters disconnected from the one containing $\mathbf{u}$. One can also see that if $\mathcal{A}$ has full rank, all points in the immediate neighborhood of $\tilde{\mathbf{w}} = \mathbf{u}$ give a learning error larger than 0, *i.e.* there is a simple minimum at $\mathbf{u}$.

Assume that the learning finds one of these valleys. A small weight decay will pick out the point in the valley with the smallest norm among all the points in the valley. In general it can not be proven that picking that solution is the best strategy. But, at least from a philosophical point of view, it seems sensible, because it is (in a loose sense) the solution with the smallest complexity—the one that Ockham would probably have chosen.

The value of a weight decay is more evident if there are small errors in the targets. In that case one can go through exactly the same line of arguments as for the linear case to show that a weight decay can improve generalization, and even with the same optimal choice (17) of λ . This is strictly true only for small errors (where the linear approximation is valid).

6 NUMERICAL EXPERIMENTS

A weight decay has been tested on the NetTalk problem [11]. In the simulations back-propagation derived from the ‘entropic error measure’ [12] with a momentum term fixed at 0.8 was used. The network had 7×26 input units, 40 hidden units and 26 output units. In all about 8400 weights. It was trained on 400 to 5000 random words from the data base of around 20.000 words, and tested on a different set of 1000 random words. The training set and test set were independent from run to run.